

Auteur: Nik Van Daele
Studierichting: Informatica
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$$\begin{aligned} & \int (3x + 5)\sqrt{x - 2} \, dx \text{ Neem } t = x - 2 \\ &= \int (3(x - 2 + 2) + 5)\sqrt{t} \, dt \\ &= \int (3(t + 2) + 5)\sqrt{t} \, dt \\ &= \int (3t + 11)\sqrt{t} \, dt \\ &= \int (3t\sqrt{t} + 11\sqrt{t}) \, dt \\ &= \int \left(3t^{\frac{3}{2}} + 11t^{\frac{1}{2}} \right) \, dt \\ &= \frac{6t^{\frac{5}{2}}}{5} + \frac{22t^{\frac{3}{2}}}{3} + c \\ &= \frac{6\sqrt{(x - 2)^5}}{5} + \frac{22\sqrt{(x - 2)^3}}{3} + c \end{aligned}$$

References